

Antibacterial Activity and Synergistic Effect of Biosynthesized AgNPs with Antibiotics Against Multidrug-Resistant Biofilm-Forming Coagulase-Negative Staphylococci Isolated from Clinical Samples

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Abstract Silver nanoparticles form promising template for designing antimicrobial agents against drug resistant pathogenic microorganisms. Thus, the development of a reliable green approach for the synthesis of nanoparticles is an important aspect of current nanotechnology research. In the present investigation, silver nanoparticles synthesized by a soil *Bacillus* sp. were characterized using UV–vis spectroscopy, FTIR, SEM, and EDS. The antibacterial potential of biosynthesized silver nanoparticles, standard antibiotics, and their conjugates were evaluated against multidrug-resistant biofilm-forming coagulase-negative *S. epidermidis* strains, *S. aureus*, *Salmonella Typhi*, *Salmonella Paratyphi*, and *V. cholerae*. Interestingly, silver nanoparticles (AgNPs) showed remarkable antibacterial activity against all the test strains with the highest activity against *S. epidermidis* strains 145 and 152. In addition, the highest synergistic effect of AgNPs was observed with chloramphenicol against *Salmonella typhi*. The results of the study clearly indicate the promising biomedical applications of biosynthesized AgNPs.

Keywords Silver nanoparticle · *Bacillus* sp. · CoNS · Biosynthesis · Synergistic activity

Introduction

Metal nanoparticle synthesis is an interesting area in nanotechnology due to the remarkable optical, magnetic, electrical, catalytic, and biomedical properties of metal nanoparticles. Unfortunately, available chemical and physical methods have various disadvantages, such as the use of toxic chemicals, high energy requirements, and difficult purification steps [1]. Biological methods of nanoparticle synthesis using microorganisms, plants, proteins,

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polypeptides, nucleic acids, etc. [2–4] can be considered as a promising alternative to these methods because of their effectiveness, flexibility, and ability to generate nanoparticles with defined size and morphology [5, 6]. Also, microbial synthesis of metal nanoparticles is considered as highly promising because of ease of culture handling and the occurrence of synthesis at ambient temperature and pressure [7, 8]. As microorganisms exhibit varying interaction with metals, the screening of novel microbial sources may lead to the identification of organisms with enhanced nanoparticle biosynthetic potential.

Silver nanoparticles (AgNPs) are well known for their amazing antimicrobial activity, and most of the biosynthesized AgNPs prove their major application as antimicrobial agents. Now, they are widely used for the disinfection of medical instruments also in medical supplies such as wound dressings, tissue scaffolds, intermittent catheters, and orthopedic prostheses [9]. Extracellularly synthesized silver nanoparticles by *Bacillus flexus* have been reported to have antibacterial effect on clinically isolated multidrug-resistant *Escherichia coli*, *Bacillus subtilis*, *Streptococcus pyogenes*, and *Pseudomonas aeruginosa* [10]. We have also demonstrated the antibacterial activity of AgNPs synthesized by a novel strain of *Pseudomonas* sp. isolated from marine source against clinically significant *Salmonella typhi*, *Vibrio cholerae*, *Bacillus subtilis*, and *Staphylococcus aureus* [11]. The enhanced antimicrobial activity of AgNPs compared to bulk silver metal is due to their high specific surface area and large fraction of surface atoms [12].

With the prevalence and increased resistance of microorganisms to multiple antibiotics, there is a need for new antimicrobial agents. Studies on antibiotics formulated with metal nanoparticles suggest the promising applications of nanoparticle-based antimicrobial formulations [7]. Recently, the enhanced antimicrobial activities of streptomycin against *E. coli*, *Salmonella typhi*, and *S. aureus* and amphotericin against *Candida albicans* in combination with biosynthesized AgNPs were reported [13]. In addition, the presence of AgNPs has also reported to enhance the antibacterial activity of ceftriaxone, carbenicillin and nitrofurantoin against *E.coli*, *K. pneumoniae* and *Proteus* sp. [14]. The silver nanoparticles synthesized by *Alternaria alternata* showed enhanced fungicidal activity with the antifungal agent fluconazole against *Phoma glomerata*, *Phoma herbarum*, *Fusarium semitectum*, *Trichoderma* sp., and *C. albicans* [15]. Very interestingly, surface applications of nanoparticle based formulations to prevent the device-associated infections caused by coagulase-negative staphylococci (CoNS) are highly promising. This is because of the emerging pathogenic role of CoNS like *Staphylococcus epidermidis*. As most of the CoNS are present as commensals, the infection is greatly considered as a result of their accidental introduction into the body which is often associated with the insertion of medical devices. Once introduced into the body, they cause the so-called difficult-to-treat condition because of their remarkable biofilm-forming property and multiple antibiotic resistance. As AgNPs execute antimicrobial activity through its effect on multiple cellular targets, the modification of medical devices with AgNPs expect to prevent CoNS growth on their surface.

In the current study, a bacterial isolate from soil was used for the intracellular synthesis of AgNPs. By molecular identification, the isolate was found to be a strain of *Bacillus* sp. The antibacterial activity of biosynthesized AgNPs was analyzed using AgNPs alone and also in combination with different antibiotics. The test organisms used were *S. aureus*, *Salmonella typhi*, *Salmonella paratyphi*, *V. cholerae*, and multidrug-resistant biofilm-forming coagulase-negative *S. epidermidis* strains 73, 145, and 152 isolated from clinical samples. Very interestingly, AgNPs and AgNPs-antibiotic combination were found to have a strong inhibitory effect on test organisms. As the AgNPs synthesized by *Bacillus* sp. or other microbial sources are least investigated for their synergistic activity with antibiotics against clinically significant CoNS, the result of the study is very significant with promising applications.

Materials and Methods

Isolation of AgNP-Synthesizing Bacteria

Soil samples collected from jewellery premises were used as sources of metal-resistant bacteria. The samples were serially diluted in sterile 0.8 % NaCl and were plated onto nutrient agar medium containing 0.5 % peptone, 0.15 % beef extract, 0.15 % yeast extract, 1 % NaCl, and 2 % agar. The colonies obtained after incubation at room temperature for 24 h were picked and purified by repeated streaking on nutrient agar and were screened for their potential to synthesize silver nanoparticles. Strain SJ 8, having the capability of AgNP synthesis, was selected for further studies.

Molecular Identification

Molecular identification of the selected isolate was carried out by 16S ribosomal DNA (rDNA) sequence-based method. For this, genomic DNA was isolated from the selected strain, and the quality of the DNA was checked by agarose gel electrophoresis. Isolated genomic DNA was used as a template for polymerase chain reaction (PCR). Forward and reverse primers used for the PCR amplification of 16S rDNA were 27F (5'-AgA gTTTgA TCM Tgg CTC-3') and 1525R (5'-AAG gAggTg WTC CAR CC-3'), respectively [11]. PCR was conducted in a 50 µl reaction volume containing 50 ng of genomic DNA, 20 pmol of each primer, 1.25 units of Taq DNA polymerase, 200 µM of each dNTP, and 1× PCR buffer. The reaction was performed for 35 cycles in a Mycycler™ (Bio-Rad, USA) with the initial denaturation at 94 °C for 3 min, cyclic denaturation at 94 °C for 30 s, annealing at 58 °C for 30 s, and extension at 72 °C for 2 min with a final extension of 7 min at 72 °C. The PCR product was analyzed by agarose gel electrophoresis. The product was then gel purified and used as a template for sequencing PCR. Sequencing PCR was carried out using the BigDye Terminator Sequence Ready Reaction Mix (Applied Biosystems). The sequencing product was purified, and the sequence run was carried out in the DNA sequencer ABI 310 Genetic Analyzer. The 16S rDNA sequence data thus obtained were aligned using the BioEdit program and subjected to BLAST analysis.

Synthesis of Silver Nanoparticles

Bacteria isolated from soil samples were inoculated into 100 mL nutrient broth (0.5 % peptone, 0.15 % beef extract, 0.15 % yeast extract, and 1 % NaCl, pH-7.0) and incubated in a rotating shaker (200 rpm) at room temperature for 24 h. After incubation, the biomass was obtained by centrifugation at 10,000 rpm for 10 min. For silver nanoparticle synthesis, about 2 g of bacterial wet biomass was resuspended in 100 mL aqueous solution of filter sterilized 1 mM AgNO₃ in a 250-mL Erlenmeyer flask. The mixture was then kept on a rotating shaker set at 200 rpm for a period of 72 h at room temperature under visible light. The bacterial biomass incubated without silver nitrate solution was maintained as control.

Purification of Silver Nanoparticles

After the color change from pale yellow to brown, the whole bacterial mixture was centrifuged at 10,000 rpm for 15 min under sterile conditions, and the collected pellets were washed and resuspended in 50 mM Tris buffer (pH 7.0). Then, the cells were disrupted by ultrasonication over three 15-s periods with an interval of 45 s between

periods. The resulting solution was filtered through a 0.22 μm filter (Millipore) to remove cell debris, and these purified AgNPs were used for further characterization [16, 17].

Characterization of Silver Nanoparticles

The biosynthesis of AgNPs was primarily observed by the color change of the biomass from pale yellow to brown in the presence of 1 mM AgNO_3 . Further characterization of AgNPs was carried out by scanning the UV–visible spectrum of the sample using Hitachi U5100 in the range of 200–800 nm at a resolution of 1 nm. The possible biomolecules that stabilize AgNPs were identified by Fourier transform infrared spectrophotometry (FTIR) at a resolution of 4 cm^{-1} in the range of 4,000–450 cm^{-1} (PerkinElmer, Spectrum 400 FTIR spectroscopy). The size and morphology of the AgNPs were characterized by scanning electron microscopy (SEM). For SEM analysis, air-dried samples were mounted on specimen stubs with double adhesive tape, coated with platinum in a sputter coater, and examined under JEOL 6390 SEM JSM at 10 kV. Also, the presence of elemental silver in the sample was confirmed by energy-dispersive spectroscopy (EDS) combined with SEM.

Antibacterial Activity Analysis of Silver Nanoparticles

The antibacterial activity of biosynthesized AgNPs was initially evaluated singly and then in combination with antibiotics by standard well diffusion and disk diffusion method in Mueller–Hinton Agar (MHA) plates as per previous reports [13, 18, 19]. Three multidrug-resistant biofilm-forming coagulase-negative *S. epidermidis* strains (73, 145, and 152) isolated from patients at the MOSC Medical College, Kolencherry, Kerala, India, were selected as major test organisms for the study. The clinical isolates *S. epidermidis* 73, 145, and 152 were isolated from the pus, catheter tips, and blood samples, respectively, and were identified by the biochemical tests. The isolates were also studied for biofilm formation both qualitatively and quantitatively (unpublished data). In addition to this, other pathogenic bacteria of the gram-positive and gram-negative groups like *S. aureus* (MTCC 87), *V. cholerae* (MTCC 3906), *Salmonella typhi*, and *Salmonella paratyphi* (collected from the clinical lab) were also used as test organisms to study the antibacterial activity of biosynthesized AgNPs.

Pure cultures of *S. epidermidis* strains were grown in trypticase soy broth, and other bacterial pathogens were grown in the nutrient broth at 37 °C for 18–24 h. Wells of 6 mm were made on the Mueller–Hinton agar plates using gel puncture, and the plates were inoculated with bacterial pathogens to create a confluent lawn of bacterial growth. Using a micropipette, 10 μL (10 μg) of the biosynthesized AgNP solution was poured onto a corresponding well as per previous report [12]. Also, 10 μL of 1 mM AgNO_3 solution and 10 μL of bacterial supernatant were poured onto respective wells as controls. The diameter of the zone of inhibition in millimeters around each well was measured after incubation at 37 °C for 24 h.

Combined Effects of Antibiotics and Biosynthesized AgNPs

The standard disk diffusion method was used to evaluate the synergistic effect of antibiotics with biosynthesized AgNPs against test strains on Mueller–Hinton agar plates. The standard antibiotic disks were (Fusidic acid-30 μg /disk, Gentamycin-10 μg /disk, Ciprofloxacin-30 μg /disk, Erythromycin-15 μg /disk, Penicillin-10 μg /disk, Chloramphenicol-50 μg /disk,

Levofloxacin - 5 µg/disk, Nalidixic acid-30 µg/disk and Ampicillin-25 µg/disc) purchased from HiMedia (Mumbai, India). To determine the combined effects, each standard antibiotic disk was impregnated with 10 µL of freshly prepared AgNPs with the final content of 10 µg of AgNPs per disk [12, 20]. The inocula were prepared by diluting the overnight cultures of bacterial pathogens in nutrient broth and clinical strains of *S. epidermidis* in trypticase soy broth with 0.9 % NaCl to a 0.5 McFarland standard and were swabbed uniformly onto the individual plates using sterile cotton swabs. Then, AgNP-impregnated antibiotic disks and antibiotic disk alone as control were placed on the surface of the plates. After incubation at 37 °C for 24 h, the different levels of zone of inhibition around each antibiotic were measured. The assays were performed in triplicate.

Results and Discussion

Surface modification of biomedical devices to minimize the introduction of CoNS into the body is an area of emerging interest. Among the various methods for surface modification, the use of AgNPs is highly promising because of its effects on multiple cellular targets which minimize the rapid evolution of resistant strains. AgNPs synthesized by diverse microorganisms show variable antimicrobial properties because of their difference in shape, size, and surface properties of nanoparticles. Therefore, screening novel microbial sources for the identification of AgNPs with enhanced activity and better stability is highly interesting. In the current study, soil samples from jewellery premises were selected as sources of bacteria. This is because of the possible adaptations expected in these organisms due to the interaction with metals. Among the bacterial isolates purified from the soil samples, SJ 8 was found to have the ability to form silver nanoparticles intracellularly as observed by the change in color of the reaction. The selected isolate was further subjected to molecular identification by the 16S rDNA sequencing-based method. The sequence data were subjected to BLAST analysis, and the result showed maximum identity of 100 % to various *Bacillus* sp., mainly *Bacillus cereus*. Thus, the isolate can be considered as a strain of *Bacillus* sp. and can be represented as *Bacillus* sp. SJ 8. The 16S rDNA sequence of the isolate was submitted to NCBI under accession number KF598859.

UV–Vis Spectroscopy Analysis

Interestingly, when the bacterial biomass was treated with 1 mM AgNO₃, it resulted in the biosynthesis of AgNPs within 12 h of incubation at room temperature. Also, the intensity of the brown color increased up to 24 h and was maintained throughout the experiment. But, no color change was observed in the flasks containing bacterial biomass without 1 mM AgNO₃ (Fig. 1). The appearance of the brown color in the silver nitrate-treated flask clearly indicates the formation of AgNPs [21, 22]. The UV–visible spectrum illustrated in Fig. 2 shows an absorption peak at 440 nm, corresponding to the surface plasmon resonance band of silver nanoparticles. The presence of such peak at this range is previously reported for biosynthesized AgNPs from *Cryphonectria* sp., *B. subtilis*, and *B. cereus* NK1 [13, 23, 24]. Surface plasmon resonance is due to the collective excitation of conduction electrons around the nanoparticle surface which display characteristic optical absorption spectra in the UV–visible region [25, 26]. As small nanoparticle exhibit only one SPR band whereas anisotropic particles show more than one band, UV vis spectroscopy analysis data is also suggested to represent morphology and stability of nanoparticles [6]. Based on this, the silver nanoparticles synthesized by the *Bacillus* sp. used in the study can be considered as small and isotropic.

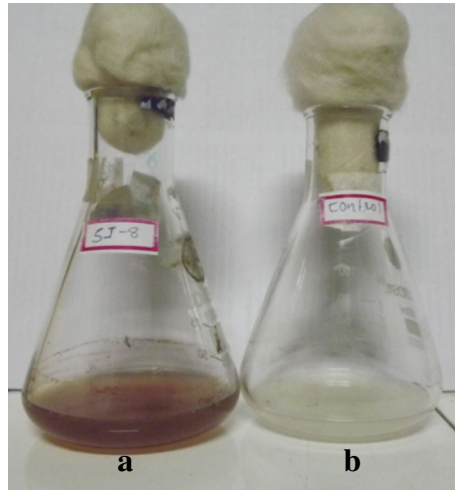


Fig. 1 Visual observation of AgNPs synthesized by *Bacillus* sp. SJ 8 after 24 h. **a** Bacterial biomass with 1 mM AgNO_3 solution (color change from pale yellow to brown). **b** Bacterial biomass without AgNO_3

FTIR Analysis

The possible biomolecules responsible for the capping and stabilization of silver nanoparticles synthesized by *Bacillus* sp. were identified by FTIR spectroscopy. IR spectroscopy is a promising tool for structural elucidation and compound identification, because different functional groups absorb characteristic frequencies of IR radiation. The FTIR spectra showed a distinct absorption peak located at $3,336$ and $1,635\text{ cm}^{-1}$ and moderate bands at $2,111$, $1,366$, and $1,217\text{ cm}^{-1}$ (Fig. 3). The strong bands at $3,336$ and $1,635\text{ cm}^{-1}$ were assigned to the

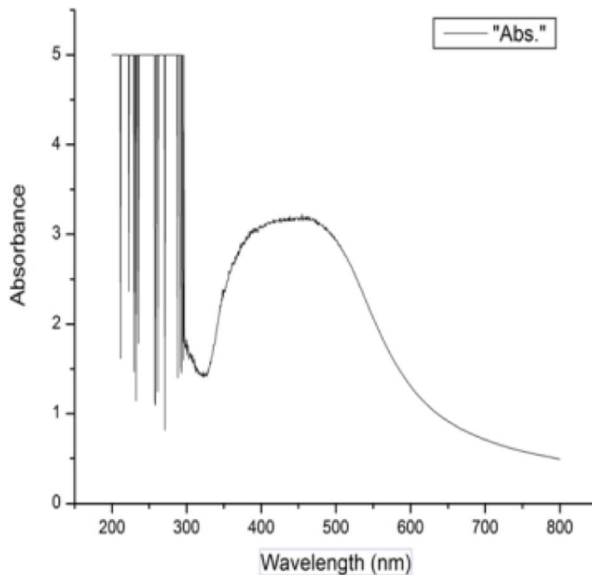


Fig. 2 The UV-vis absorption spectrum of silver nanoparticles synthesized by *Bacillus* sp. SJ 8. The absorption spectrum of silver nanoparticles exhibited a peak at 440 nm

stretching and bending vibrations of the primary amines, respectively, while small bands at $1,366\text{ cm}^{-1}$ were due to the stretching vibrations of aromatic amines. It was also reported that proteins can bind to nanoparticles either through free amine groups or cysteine residues [14, 26]. Stabilization of silver nanoparticles by the surface bound protein might have occurred during their synthesis by microbes as observed in the FTIR analysis. So, reduction of silver ions to silver nanoparticles might have assisted by subsequent stabilization during the microbial synthesis.

SEM–EDS Studies

The size and shape of purified silver nanoparticles were elucidated using SEM. The SEM image confirmed the presence of sphere-shaped silver nanoparticles, and their sizes were found to be in the range of 14–42 nm (Fig. 4). The presence of elemental silver in the sample was confirmed by EDS as a result of strong signal at 3 keV (Fig. 5). An optical absorption band at 3 keV is the characteristic feature of silver nanocrystallites due to surface plasmon resonance [27–29]. An additional peak corresponding to Al was also observed due to the aluminum foil on which the sample was prepared for SEM and EDS.

Antibacterial Activity of AgNPs

The antibacterial effect of biologically synthesized nanoparticles was tested against a panel of human pathogens comprising both gram-positive and gram-negative bacteria by well diffusion assay. Clinical isolates of CoNS were used in the study to explore the possible applications of antibiotic-AgNP combinations to the surface of medical devices to minimize the CoNS infection. Other pathogens were selected to study the broad antimicrobial effect of the AgNP combination. Interestingly, AgNPs at $10\text{ }\mu\text{g}$ showed remarkable antibacterial activity against *S. epidermidis* strains 73, 145, and 152 and also against *S. aureus*, *Salmonella typhi*, *Salmonella paratyphi*, and *V. cholerae*. The diameter of the zone of inhibition of AgNPs against test all test strains is shown in Table 1. The AgNPs showed more activity than AgNO_3 , and there was no zone of inhibition by the bacterial supernatant which confirmed that the

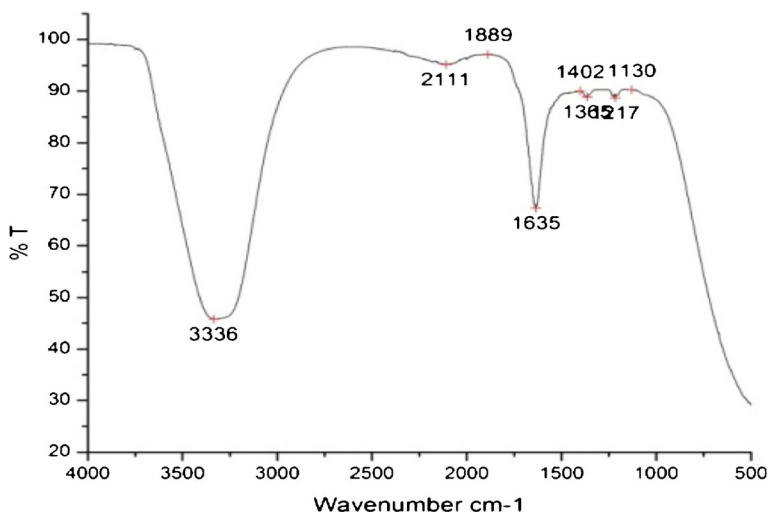


Fig. 3 FTIR spectrum of AgNPs synthesized by the reduction of AgNO_3 by *Bacillus* sp.

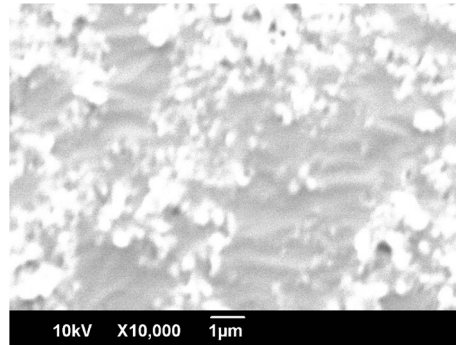


Fig. 4 Scanning electron microscopic image of AgNPs synthesized by *Bacillus sp.* SJ 8

activity was due to the synthesized AgNPs only. Also, the highest antibacterial activity was observed against *S. epidermidis* strains 145 and 152, and the result is highly supportive of the possible use of the biosynthesized AgNPs for the surface modification of medical devices. The coating of these AgNPs on medical devices may prevent the colonization and proliferation of opportunistic pathogens like CoNS which are associated with infections in hospitalized patients and outpatients. In addition, biosynthesized AgNPs showed a broad spectrum of antibacterial activity against other test strains, and the result is highly promising in the field of nanomedicine. There are various reports on the antibacterial effect of silver nanoparticles. Silver nanoparticles synthesized extracellularly by the culture supernatants of *Bacillus megaterium* (NCIM 2326) have been found to have antibacterial activity against *Streptococcus pneumoniae* and *Salmonella typhi* [19]. Also, AgNPs biosynthesized by *S. aureus* have been proved to have excellent antibacterial activity against methicillin-resistant staphylococci [18]. Very importantly, biosynthesized AgNPs have been reported to have superior bactericidal activity to chemically synthesized colloidal-Ag nanoparticles [30]. The mode of action of AgNPs against pathogenic bacteria may involve inhibition of bacterial enzymes by reacting with thiol groups and depletion of cellular antioxidant capacity [31, 32]. The role of silver in generating free radicals with damaging effect to the bacterial cell

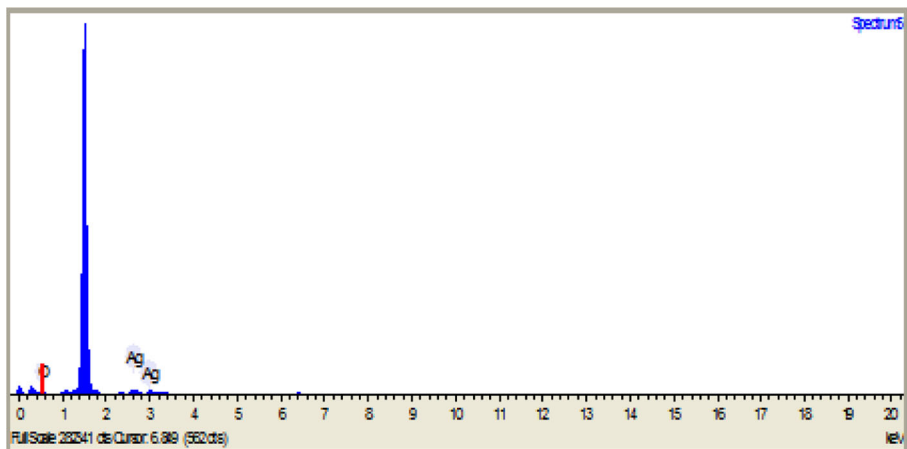


Fig. 5 EDS analysis of silver nanoparticles

Table 1 Diameter of the zone of inhibition of biosynthesized AgNPs against pathogenic bacteria

Bacteria	Zone of inhibition (mm)		
	Biosynthesized AgNPs (10 μ L)	Control A (1 mM AgNO ₃)	Control B (bacterial supernatant)
<i>S. epidermidis</i> strain 73	15	10	No zone
<i>S. epidermidis</i> strain 145	19	8	No zone
<i>S. epidermidis</i> strain 152	19	12	No zone
<i>S. aureus</i>	18	No zone	No zone
<i>Salmonella typhi</i>	13	10	No zone
<i>Salmonella paratyphi</i>	15	No zone	No zone
<i>V. cholerae</i>	18	8	No zone

wall has also been suggested [33, 34]. In addition, silver nanoparticles are also suggested to interact with bacterial DNA and create problems in DNA replication, leading to cell death [35].

Synergistic Effect of AgNPs with Antibiotics

The combination of biosynthesized AgNPs and antibiotics was investigated against all the test strains by disk diffusion method. Table 2 shows the diameter of the zone of inhibition around different antibiotic disks with and without silver nanoparticles against the test strains. The antibacterial activities of all antibiotics except fusidic acid against *S. epidermidis* strain 73 and ciprofloxacin against *S. epidermidis* strain 152 were increased against all the test strains

Table 2 Zone of inhibition in millimeters of different antibiotics with and without AgNPs against gram-positive and gram-negative bacteria

Bacteria	Antibiotics	Zone of inhibition (mm)		Increase in fold area ($b^2 - a^2/a^2$)
		Antibiotic only (a)	Antibiotic+AgNPs (b)	
<i>S. epidermidis</i> strain 73	Fusidic acid	25	25	0
	Gentamycin	14	15	0.15
<i>S. epidermidis</i> strain 145	Gentamycin	7	10	1.04
	Ciprofloxacin	22	23	0.09
<i>S. epidermidis</i> strain 152	Fusidic acid	10	13	0.7
	Ciprofloxacin	18	18	0
<i>S. aureus</i>	Erythromycin	10	15	1.25
	Penicillin	No zone	14	4.4
<i>Salmonella typhi</i>	Chloramphenicol	13	20	1.37
	Levofloxacin	20	25	0.56
<i>Salmonella paratyphi</i>	Levofloxacin	15	20	0.8
	Nalidixic acid	15	16	0.14
<i>V. cholerae</i>	Erythromycin	No zone	10	1.8
	Ampicillin	No zone	12	3

In the absence of bacterial growth inhibition zones, the antibiotic disk diameter (6 mm) was used to calculate the increase in fold area

(Fig. 6). The highest fold increase in area was observed for penicillin against *S. aureus*. But highest synergistic effect was observed with chloramphenicol-AgNPs combination against *S. typhi*. Very interestingly, this result showed that the pathogenic bacteria which were completely resistant to antibiotics become susceptible to antibiotics in combination with AgNPs. Thus, this result provided helpful insight into the development of new antimicrobial agents. The enhanced antibacterial activities of commonly used antibiotics such as methicillin, chloramphenicol, and ciprofloxacin in the presence of biosynthesized AgNPs against *S. aureus* MTCC96, *S. pyogenes* MTCC1925, *Salmonella enterica* MTCC735, and *Enterococcus faecalis* MTCC2729 were previously reported [36]. Extracellular synthesis of AgNPs using *Trichoderma viridae* and the combined effects of biosynthesized AgNPs with antibiotics have been also studied. They explained the potential of AgNPs to enhance the antibacterial activity of ampicillin, kanamycin, erythromycin, and chloramphenicol against *E. coli*, *Salmonella typhi*, *S. aureus*, and *Micrococcus luteus* [20]. They also suggested the possible chelation between antibiotics and AgNPs through the active hydroxyl and amido groups present in the antibiotic molecules. And, this binding may be one of the reasons behind their synergistic

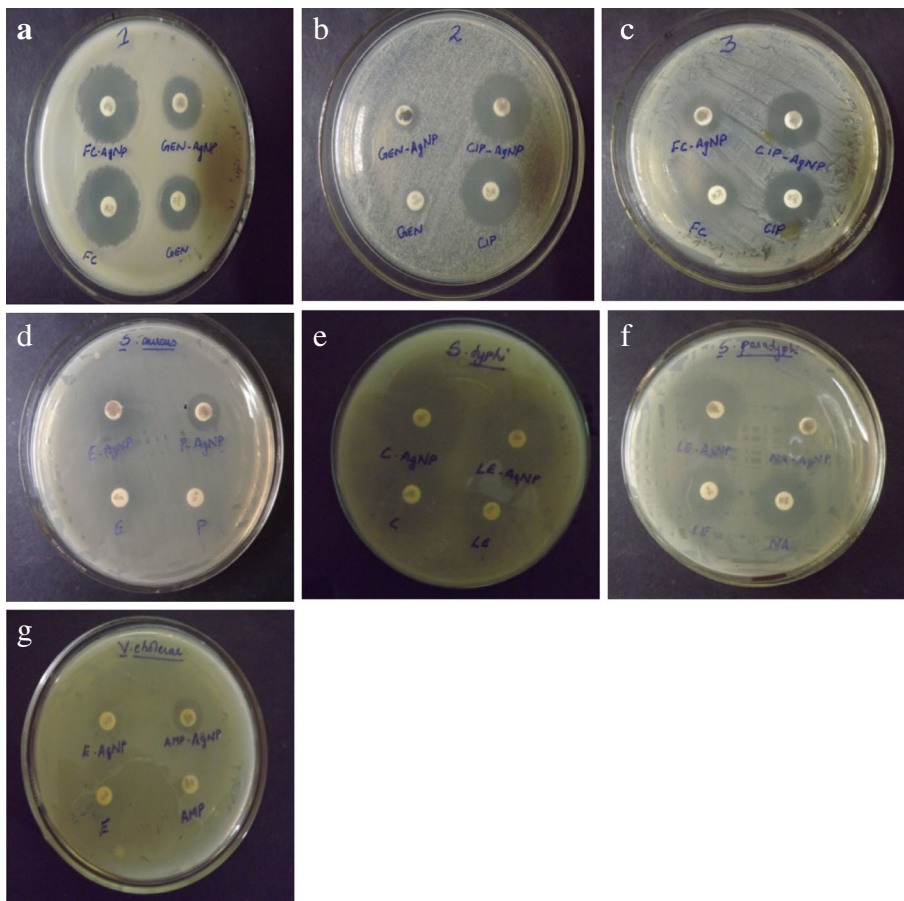


Fig. 6 Synergistic activity of biosynthesized AgNPs with antibiotics against **a** *S. epidermidis* strain 73, **b** *S. epidermidis* strain 145, **c** *S. epidermidis* strain 152, **d** *Staphylococcus aureus*, **e** *Salmonella typhi*, **f** *Salmonella paratyphi*, and **g** *Vibrio cholerae*

activity. Thus, the ability of AgNPs in combination with antibiotics to act against those organisms which were resistant to the same antibiotics provides new opportunities for the management of the pathogenic organisms. The effect of the antibiotic–AgNP combination towards multiple cellular targets can be highly effective in managing the emerging pathogens. Infections caused by multidrug-resistant pathogens demand the development of treatment strategies which are effective, less toxic, and economical. Thus, our results suggest biosynthesized AgNPs as potential broad-spectrum agents against a variety of multidrug-resistant bacteria and it have direct applications as surface modification agents on medical devices or pharmaceutical products. This may help to prevent transmission of drug-resistant pathogenic bacteria in different clinical environments.

Conclusion

Metal nanoparticles are known to have emerging applications, and microbial synthesis will be a promising method to generate nanoparticles with promising applications. In the present study, the silver nanoparticles synthesized using *Bacillus* sp. were characterized by UV–vis spectroscopy, FTIR, SEM, and EDS. Also, biosynthesized AgNPs showed promising antibacterial and synergistic activity with antibiotics against multidrug-resistant biofilm-forming coagulase-negative *S. epidermidis* strains and other pathogenic bacteria. The biosynthesized silver nanoparticles as explained in the study can have immediate application in formulating silver-based dressings, silver-coated medical devices, etc. to manage multidrug-resistant pathogenic microorganisms. The highlights of the study are the environment-friendly method for the synthesis of AgNPs and its promising activity against pathogens like CoNS, which cause device-related infections.

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